Samet Kocbay

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Winthirstrasse 16, Munich - 80639, Germany

Professional experience

- **Max Planck Institute for Plasma Physics []** *Since October 2025*
  Working Student/ Thesis
  - Deriving a self-consistent coupling for electromagnetic interactions of a fusion reactor.
- **BMW []** *October 2024 - September 2025*
  Working student
  - Developed a Gaussian process surrogate model for acoustic analysis.
  - Built and deployed a neural network to predict optimal solver settings.
- **Compact Dynamics []** *March 2024 - September 2024*
  Bachelor thesis
  - Developed an RNN for real-time rotor temperature prediction in electric motors.
  - Validated the model on the test bench.
- **Institute of Fluid Mechanics []** *July 2023 – December 2023*
  Student Assistant
  - Researched and implemented a physics-informed neural network for inverse Navier–Stokes problems.
  - Analyzed the influence of additional loss function terms.

Education

- **M.Sc. Computational Science and Engineering** *Since October 2024*
  Technical University of Munich (TUM)
- **B.Sc. Mechanical Engineering** *October 2019 - September 2024*
  Karlsruhe Institute of Technology (KIT)

Projects

- **KA RaceIng: Aerodynamics** *October 2021 - September 2022*
  - Designed Rear-wing geometry integrated with autonomous sensor housing.
  - Analyzed complex fan-induced flows using the Moving Reference Frame method.
- **KA RaceIng: Monocoque and FEM** *October 2021 - September 2022*
  - Optimized carbon monocoque structure through multi-stage optimization, balancing mass efficiency with structural integrity.
  - Enhanced manufacturing precision by implementing a laser projection system for optimized ply placement in chassis molds.

Leadership Experience

- **Formula Student: Head of Aerodynamics** *September 2022 - October 2023*
  - Led a team of 13 engineers, overseeing aerodynamic design, lightweight structures, and CFD analysis.
  - Maintained team motivation and efficiency in a high-pressure environment.

Skills

- **Languages:** German (Native), English (C1/TOEFL: 111/120), Turkish
- **Programming Languages:** Python, C++, MATLAB, JAX
- **Data Science & Machine Learning:** PyTorch, GPyTorch
- **CAD & CAE:** Altair Hyperworks, Ansys Mechanical, STAR-CCM+, OpenFOAM, Ansa, PTC Creo